



Cyber Security & Digital Forensics

CSE - 4207

Lecture - 03

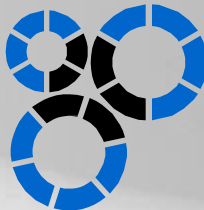
Dr. Md. Alam Hossain

Professor

Jashore University of Science & Technology

Outline

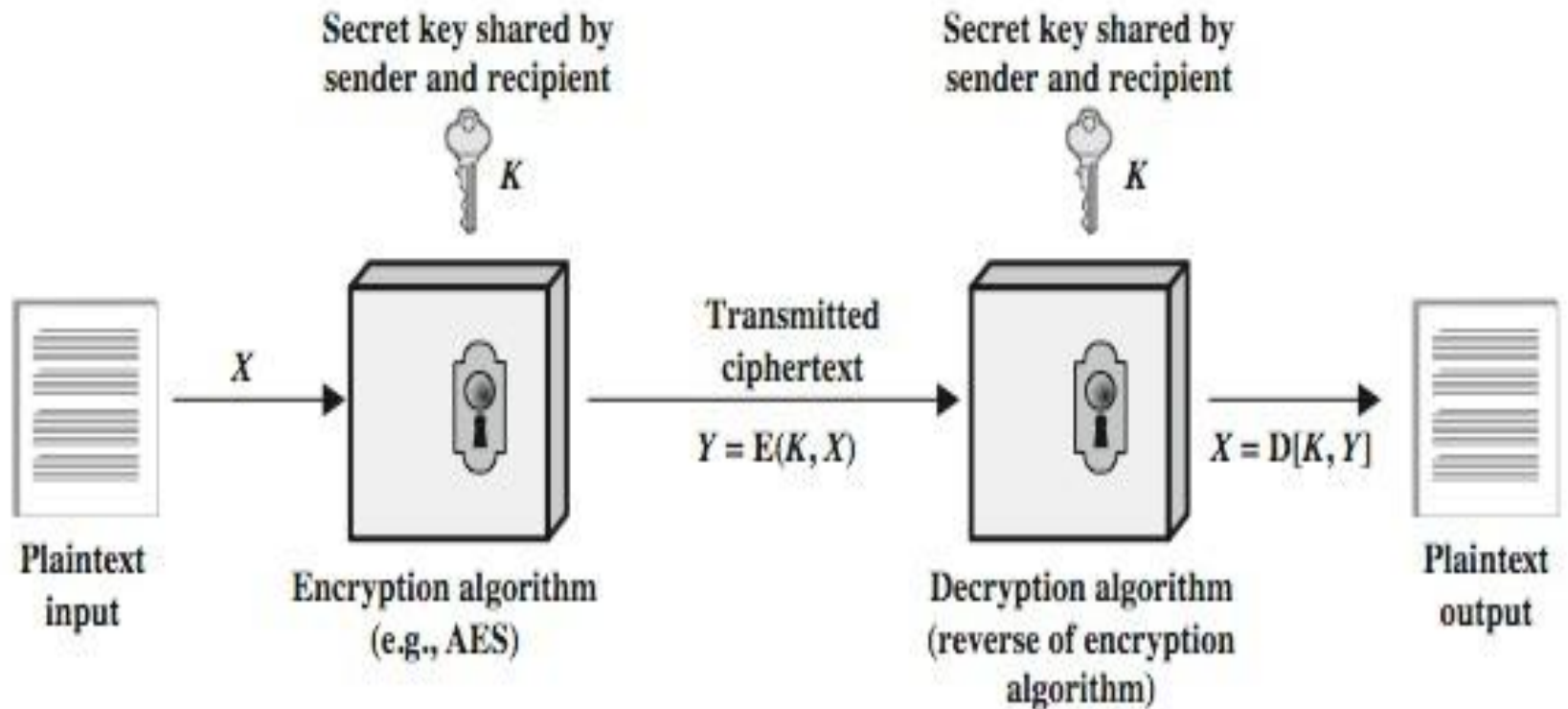
- Some Basic Terminology
- Symmetric Cipher Model
- Cryptanalysis
- Cryptanalytic Attacks
- Brute Force Search
- Monoalphabetic Cipher
- Caesar Cipher
- Language Redundancy and Cryptanalysis
- English Letter Frequencies
- Example Cryptanalysis



◉ Some Basic Terminology

- Plaintext- original message
- Ciphertext- coded message
- Cipher- algorithm for transforming plaintext to ciphertext
- Key- info used in cipher known only to sender/receiver
- Encipher (encrypt)- converting plaintext to ciphertext
- Decipher (decrypt)- recovering ciphertext from plaintext
- Cryptography- study of encryption principles/methods
- Cryptanalysis (code breaking)- study of principles/ methods of deciphering ciphertext without knowing *key*
- Cryptology-field of both cryptography and cryptanalysis

◉ Symmetric Cipher Model



⦿ Requirements

- two requirements for secure use of symmetric encryption:
 - a strong encryption algorithm
 - a secret key known only to sender / receiver
- mathematically have:
- $Y = E(K, X)$
- $X = D(K, Y)$
 - assume encryption algorithm is known
 - implies a secure channel to distribute key

Cryptography

- Can characterize cryptographic system by:
 - type of encryption operations used
 - Substitution
 - Transposition
 - Product
 - number of keys used
 - Single-key or private
 - Two-key or public
 - way in which plaintext is processed
 - Block
 - Stream

◉ Cryptanalysis

- objective to recover key not just message
- general approaches:
 - cryptanalytic attack
 - brute-force attack
- if either succeed all key use compromised

⦿ Cryptanalytic Attacks

- ciphertext only
 - only know algorithm & ciphertext, is statistical, know or can identify plaintext
- known plaintext
 - know/suspect plaintext & ciphertext
- chosen plaintext
 - select plaintext and obtain ciphertext
- chosen ciphertext
 - select ciphertext and obtain plaintext
- chosen text
 - select plaintext or ciphertext to en/decrypt

◉ More Definitions

- Unconditional security
 - no matter how much computer power or time is available, the cipher cannot be broken since the ciphertext provides insufficient information to uniquely determine the corresponding plaintext
- Computational security
 - given limited computing resources (i.e. time needed for calculations is greater than age of universe), the cipher cannot be broken

◉ Brute Force Search

- always possible to simply try every key
- most basic attack, proportional to key size
- assume either know / recognize plaintext

Classical Substitution Ciphers

- where letters of plaintext are replaced by other letters or by numbers or symbols
- or if plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns

◉ Monoalphabetic Cipher

- rather than just shifting the alphabet
- could shuffle (jumble) the letters arbitrarily
- each plaintext letter maps to a different random ciphertext letter
- hence key is 26 letters long

Plain: abcdefghijklmnopqrstuvwxyz

Cipher: DKVQFIBJWPESCXHTMYAUOLRGZN

Plaintext: ifwewishtoreplaceletters

Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA

◉ Monoalphabetic Cipher Security

- now have a total of $26! = 4 \times 10^{26}$ keys
- with so many keys, might think is secure
- but would be **!!!WRONG!!!**
- problem is language characteristics

⦿ Caesar Cipher

- Earliest known substitution cipher - by Julius Caesar
- First attested use in military affairs
- Replaces each letter by 3rd letter on
- example:
 - meet me after the toga party
 - PHHW PH DIWHU WKH WRJD SDUWB

Caesar Cipher

- can define transformation as:

a b c d e f g h i j k l m n o p q r s t u v w x y z

D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

- mathematically give each letter a number

a b c d e f g h i j k l m n o p q r s t u v w x y z

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23 24 25

- then have Caesar cipher as:

$$c = E(k, p) = (p + k) \bmod (26)$$

$$p = D(k, c) = (c - k) \bmod (26)$$

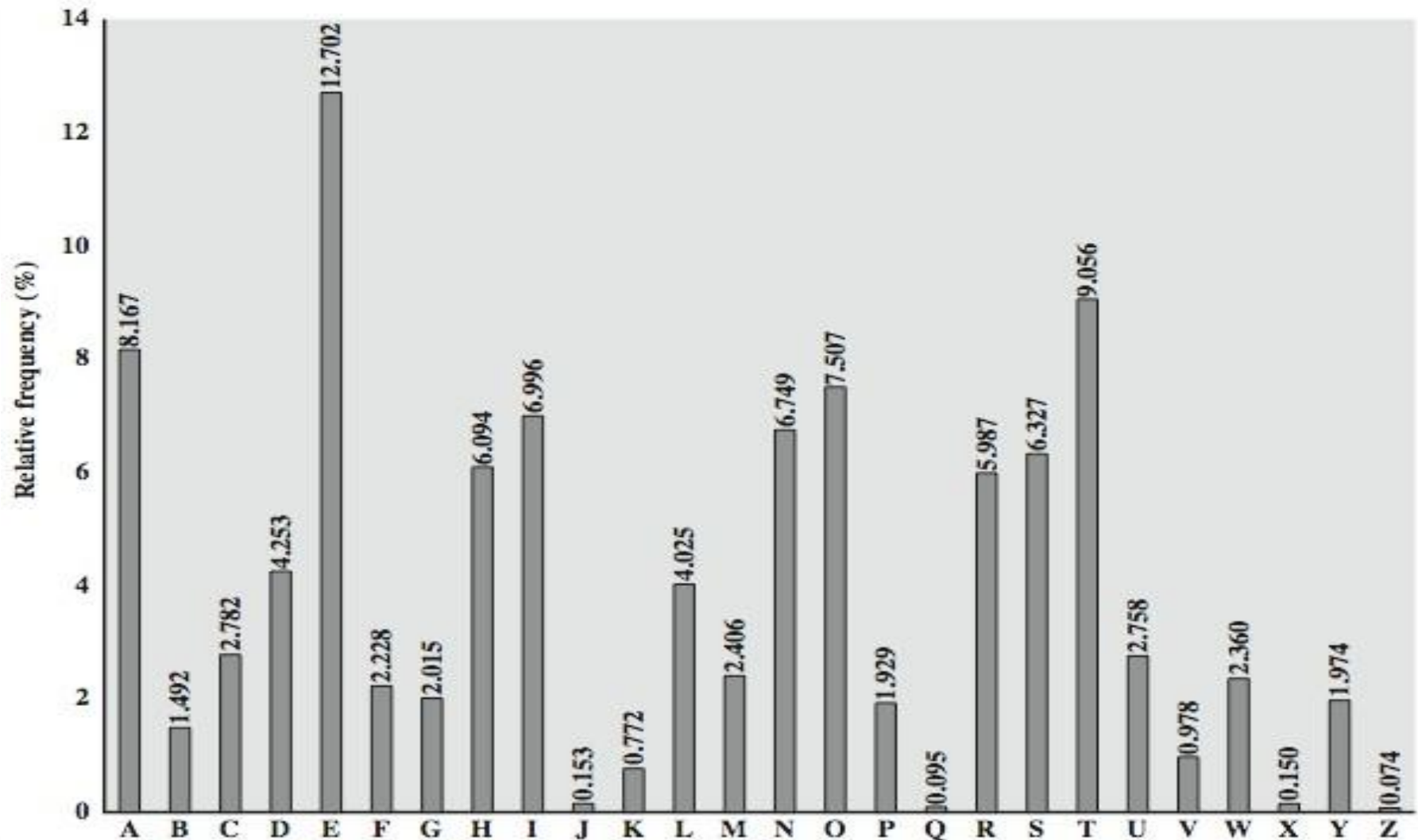
🌀 Cryptanalysis of Caesar Cipher

- only have 26 possible ciphers
 - A maps to A,B,..Z
- could simply try each in turn
- a brute force search
- given ciphertext, just try all shifts of letters
- do need to recognize when have plaintext
- eg. break ciphertext "GCUA VQ DTGCM"

Language Redundancy and Cryptanalysis

- human languages are **redundant**
- eg "th lrd s m shphrd shll nt wnt"
- letters are not equally commonly used
- in English E is by far the most common letter
 - followed by T,R,N,I,O,A,S
- other letters like Z,J,K,Q,X are fairly rare
- have tables of single, double & triple letter frequencies for various languages

English Letter Frequencies



◉ Use in Cryptanalysis

- key concept- monoalphabetic substitution ciphers do not change relative letter frequencies
- discovered by Arabian scientists in 9th century
- calculate letter frequencies for ciphertext
- compare counts/plots against known values
- if caesar cipher look for common peaks/troughs
 - peaks at: A-E-I triple, NO pair, RST triple
 - troughs at: JK, X-Z
- for monoalphabetic must identify each letter
 - tables of common double/triple letters help

◉ Example Cryptanalysis

- given ciphertext:
 - UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETS
XAIZVUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHS
XEPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ
- count relative letter frequencies (see text)
- guess P & Z are e and t
- guess ZW is th and hence ZWP is the
- proceeding with trial and error finally get:
 - it was disclosed yesterday that several informal but direct contacts have been made with political representatives of the viet cong in moscow



Thank You

